

Matrix Theory as the Engine of Big-Data Machine Learning: A Computational Companion

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Abstract

Nearly every algorithm that makes machine learning feasible at scale is a theorem of matrix analysis in an engineering costume. The Eckart-Young-Mirsky theorem underwrites all low-rank compression; the Johnson-Lindenstrauss lemma licenses dimensionality reduction independent of the ambient dimension; subspace embeddings turn tall regressions into small ones; Cheeger's inequality converts Laplacian eigenvectors into clustering algorithms; Perron-Frobenius theory sets the (size-independent) runtime of PageRank; and the spectral decay of Gram matrices makes kernel methods tractable through Nyström approximation. This expository paper presents the chain alongside an open-source Python package, `matrix-ml`, in which each theorem is paired with the computation it governs, so bounds can be verified numerically--often with equality. We report experiments in which the randomized SVD of Halko-Martinsson-Tropp matches the optimal Eckart-Young error to a relative accuracy of 10^{-5} at a $26\times$ speedup over the full SVD; sketched least squares attains a residual within 1.1% of optimal from a 2% sample of rows; power iteration converges at exactly its theoretical rate $|\lambda_2/\lambda_1|$; and PageRank converges in 34-37 iterations regardless of graph size, as the damping bound predicts. The aim is pedagogical: to make visible that the guarantees of large-scale learning *are* matrix theory.

1 Introduction

The practical questions of big-data machine learning--How can a matrix too large for memory be factorized? How can a million-dimensional dataset be compressed without destroying its geometry? How can a regression over 10^5 observations be solved from a sketch? Why does Google's eigenvector computation terminate?--all receive their answers from classical and modern matrix analysis. The purpose of this paper and its accompanying software is to line the answers up next to the theorems that produce them.

Our organizing chain is:

Eckart-Young $\longrightarrow$ randomized SVD $\longrightarrow$ Johnson-Lindenstrauss $\longrightarrow$ sketched regression
 $\longrightarrow$ PCA $\longrightarrow$ spectral clustering $\longrightarrow$ power iteration / PageRank $\longrightarrow$ Nyström.

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Sections 2–6 present the theory; Section 7 describes the package and experiments; Section 8 closes with limitations and extensions. Standard references are Trefethen–Bau [3] for the deterministic theory, Halko–Martinsson–Tropp [11] and Woodruff [13] for the randomized theory, von Luxburg [10] for spectral clustering, and Vershynin [14] for the probabilistic tools.

2 Optimal low rank and its randomized computation

Let $A \in \mathbb{R}^{m \times n}$ have singular value decomposition $A = U\Sigma V^T$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$, and let $A_k = U_k \Sigma_k V_k^T$ denote the truncation to the top k terms.

Theorem 2.1 (Eckart–Young–Mirsky [1]). *For every matrix B with $\text{rank } B \leq k$,*

$$\|A - B\|_2 \geq \sigma_{k+1}, \quad \|A - B\|_F \geq \left(\sum_{i>k} \sigma_i^2 \right)^{1/2},$$

and both bounds are attained by $B = A_k$.

Theorem 2.1 is the charter of all low-rank machine learning: whatever compression, denoising, or latent-factor structure one seeks, the truncated SVD is provably the best rank- k answer, and the singular value tail is the exact price of compression.

The full SVD costs $O(mn \min(m, n))$, unaffordable for big data. Randomized numerical linear algebra computes A_k approximately from a *sketch*.

Theorem 2.2 (Halko–Martinsson–Tropp [11]). *Let $\Omega \in \mathbb{R}^{n \times (k+p)}$ be standard Gaussian with oversampling $p \geq 2$, let Q be an orthonormal basis of the range of $(AA^T)^q A \Omega$ (q power iterations). Then*

$$\mathbb{E} \|A - QQ^T A\|_2 \leq \left[\left(1 + \frac{\sqrt{k}}{\sqrt{p-1}}\right) \sigma_{k+1} + \frac{e\sqrt{k+p}}{p} \left(\sum_{i>k} \sigma_i^2\right)^{1/2} \right]^{1/(2q+1)} \cdot \sigma_{k+1}^{2q/(2q+1)}.$$

In particular, with $p \approx 10$ and $q = 2$ the error is essentially the optimal σ_{k+1} of Theorem 2.1, at cost $O(mn(k+p))$ plus dense work on $(k+p)$ -sized matrices.

The algorithm is five lines: sketch $Y = A\Omega$, orthonormalize $Q = \text{qr}(Y)$, optionally power-iterate, form the small matrix $B = Q^T A$, and take its dense SVD. Section 7 reports that on a 3000×1500 matrix this matches the singular values of the full SVD to 1.8×10^{-5} while running $26 \times$ faster.

3 Random projections and sketched regression

Theorem 3.1 (Johnson–Lindenstrauss [2]). *Let $x_1, \dots, x_n \in \mathbb{R}^d$ and $0 < \varepsilon < 1$. If $m \geq 8 \ln(n)/\varepsilon^2$ and $S \in \mathbb{R}^{m \times d}$ has i.i.d. $N(0, 1/m)$ entries, then with high probability*

$$(1 - \varepsilon) \|x_i - x_j\|^2 \leq \|Sx_i - Sx_j\|^2 \leq (1 + \varepsilon) \|x_i - x_j\|^2 \quad \text{for all } i, j.$$

The target dimension m does not involve d : two hundred points in $\mathbb{R}^{10,000}$ embed into $\mathbb{R}^{679}$ with distortion $\varepsilon = 0.25$. Sparse constructions such as CountSketch [12] achieve the same guarantee in input-sparsity time.

For least squares $\min_x \|Ax - b\|$ with $A \in \mathbb{R}^{n \times d}$, $n \gg d$, a *subspace embedding*--an S preserving the norms of all vectors in the $(d+1)$ -dimensional subspace $\text{range}([A \ b])$ --suffices,

and Theorem 3.1-type matrices with $m = O(d/\varepsilon^2)$ rows provide one [9, 13]. The sketched solution $\tilde{x} = \arg \min_x \|SAx - Sb\|$ then satisfies

$$\|A\tilde{x} - b\| \leq (1 + \varepsilon) \min_x \|Ax - b\|. \quad (1)$$

The big regression is replaced by a small one whose size is set by d and ε , not by n .

4 PCA and spectral clustering

PCA is Theorem 2.1 applied to data. If the rows of $X \in \mathbb{R}^{n \times d}$ are centered, the SVD $X = U\Sigma V^T$ diagonalizes the covariance $C = X^T X / (n - 1) = V \text{diag}(\sigma_i^2 / (n - 1)) V^T$; the top- k right singular vectors are the principal directions, the ratios $\sigma_i^2 / \sum_j \sigma_j^2$ are the explained variances, and Eckart-Young certifies the projection as the best k -dimensional reconstruction in Frobenius norm. The singular value spectrum is the intrinsic-dimension diagnostic of the dataset.

Laplacian eigenvectors find clusters. For a graph with adjacency W and degrees D , the normalized Laplacian is $L_{\text{sym}} = I - D^{-1/2} W D^{-1/2} \succeq 0$. The multiplicity of the eigenvalue 0 counts connected components, and the second eigenvalue λ_2 (algebraic connectivity) controls the best sparse cut through Cheeger's inequality

$$\frac{\lambda_2}{2} \leq h(G) \leq \sqrt{2\lambda_2}, \quad (2)$$

where $h(G)$ is the conductance. Spectral clustering [5, 6, 10] embeds vertices by the bottom eigenvectors and clusters in that embedding; for two balanced communities the *signs of the Fiedler vector* suffice. On the stochastic block model with $n = 600$, intra-community edge probability $p = 0.10$ and inter-community probability $q = 0.02$, the Fiedler-sign partition recovers the hidden communities with 100% accuracy (Section 7).

5 Power iteration and PageRank

Power iteration--repeat $x \leftarrow Ax / \|Ax\|$ --converges to the dominant eigenvector at the geometric rate $|\lambda_2/\lambda_1|^t$; the proof is one line in the eigenbasis. PageRank [4] is power iteration on the Google matrix

$$G = \alpha P + (1 - \alpha) \frac{1}{n} J, \quad \alpha \approx 0.85, \quad (3)$$

with P the column-stochastic link matrix and J the all-ones matrix. Perron-Frobenius theory gives a unique positive stationary vector, and the damping forces $|\lambda_2(G)| \leq \alpha$ [8]; hence convergence to tolerance τ takes at most $\log \tau / \log \alpha$ iterations *independently* of n . For $\tau = 10^{-10}$ and $\alpha = 0.85$ this bound is 142; our experiments observe 34-37 iterations on random graphs of 100 to 3000 nodes. Matrix analysis, not data size, sets the runtime of one of the largest computations ever run.

6 Nyström kernel approximation

Kernel methods require the $n \times n$ Gram matrix $K_{ij} = k(x_i, x_j)$ --quadratic memory and cubic solves. The Nyström method [7] chooses $m \ll n$ landmarks, forms the blocks $C \in \mathbb{R}^{n \times m}$ and $W \in \mathbb{R}^{m \times m}$ of K , and uses

$$K \approx CW^+C^T, \quad (4)$$

a rank- m positive semidefinite approximation computable in $O(nm^2)$. Its quality is governed by the spectral decay of K --the Eckart-Young philosophy transplanted to feature space. On a clustered dataset of $n = 600$ points the relative Frobenius error of (4) falls from 0.44 at $m = 10$ landmarks to 0.05 at $m = 200$ (Table 1).

7 The computational companion

7.1 Package design

The package matrix-ml (Python, NumPy only, MIT licence) mirrors the chain module by module: lowrank (truncated and randomized SVD, Eckart-Young error audit), sketching (Gaussian and CountSketch transforms, JL distortion, sketched least squares), pca, spectral (stochastic block models, normalized Laplacians, Fiedler vectors), power (power iteration with tracked error history, PageRank), and nystrom. Twelve unit tests bind code to theorems: no random rank- k matrix ever beats the truncated SVD; a disconnected graph has $\lambda_2 = 0$ exactly; Nyström with all points as landmarks reproduces K to machine precision; the empirical power-iteration rate matches $|\lambda_2/\lambda_1|$.

The randomized SVD, the workhorse, is essentially:

```
Omega = rng.standard_normal((n, k + p))
Q, _ = np.linalg.qr(A @ Omega) # sketch the range
for _ in range(q): # power iterations
    Q, _ = np.linalg.qr(A.T @ Q)
    Q, _ = np.linalg.qr(A @ Q)
B = Q.T @ A # small (k+p) x n
Ub, s, Vt = np.linalg.svd(B) # dense SVD of the sketch
U = Q @ Ub
```

7.2 Experiments

All numbers below are produced by the four example scripts in the repository; total runtime is under half a minute on a laptop.

Eckart-Young and randomized SVD. On a 3000×1500 matrix with spectrum $\sigma_i = i^{-1.5}$ and $k = 20$, the truncated SVD achieves the optimal errors of Theorem 2.1 to eight significant digits (spectral: 1.039133×10^{-2} optimal and achieved; Frobenius: 3.430195×10^{-2} optimal and achieved). The randomized SVD with $p = 10$, $q = 2$ runs in 0.102s against 2.662s for the full SVD--a $26 \times$ speedup--with relative Frobenius error 3.130×10^{-2} versus the optimal 3.129×10^{-2} and maximal singular value deviation 1.77×10^{-5} .

Johnson-Lindenstrauss and sketched regression. Two hundred Gaussian points in $\mathbb{R}^{10,000}$ are projected to the JL dimension $m = 679$ for $\varepsilon = 0.25$; the worst pairwise distortion observed is 0.2519—sitting exactly at the shoulder of the high-probability guarantee, a reminder that Theorem 3.1 is probabilistic, not deterministic. For least squares with $n = 100,000$, $d = 50$, sketching to m rows yields residual ratios $\|A\tilde{x} - b\|/\|Ax^* - b\|$ of 1.048, 1.023, and 1.011 for $m = 500, 1000, 2000$: the $(1+\varepsilon)$ guarantee (1), realized from a 2% sample of the data.

PCA and intrinsic dimension. For 2000 points on a noisy 5-dimensional subspace of $\mathbb{R}^{300}$, PCA explains 67.1% of variance at $k = 2$, 89.2% at $k = 5$, and only 89.5% at $k = 10$: the spectrum (448.8, 357.8, 261.2, 178.4, 92.3, then noise floor) identifies the intrinsic dimension at a glance.

Spectral clustering, power iteration, PageRank, Nyström. On the SBM described in Section 4 the algebraic connectivity is $\lambda_2 = 0.3169$ and Fiedler-sign clustering attains 100% accuracy. Power iteration on a matrix with spectrum $(1, 0.9, 0.5, \dots)$ converges at empirical rate 0.9000 against the theoretical $|\lambda_2/\lambda_1| = 0.9$. PageRank with $\alpha = 0.85$, $\tau = 10^{-10}$ converges in 34, 37, and 37 iterations on random directed graphs of 100, 1000, and 3000 nodes, well under the size-independent bound of 142. Nyström errors are in Table 1.

landmarks m	10	25	50	100	200
$\ K - CW^+C^T\ _F/\ K\ _F$	0.4396	0.2446	0.1532	0.0954	0.0496

Table 1: Nyström approximation of the RBF Gram matrix of $n = 600$ clustered points in $\mathbb{R}^{20}$. The error decays with the number of landmarks at the rate set by the spectral decay of K : low-rank structure in feature space is what makes kernel methods scale.

8 Discussion

What the computations do and do not show. Finite experiments verify, they do not prove. What they provide here is precision of a kind rare in empirical work: several of the bounds hold *with equality* (Eckart-Young by construction; the power iteration rate to four digits), several are attained at their stated shoulders (JL distortion 0.2519 against $\varepsilon = 0.25$), and the rest hold with the predicted margins. Watching a bound bind is the fastest way to understand why it is the truth of the matter and not an artifact of proof technique.

Extensions. Natural next modules include leverage-score sampling and the column-subset selection problem, sketched ridge regression and kernel ridge with Nyström preconditioning, Lanczos and block-Krylov methods as the deterministic counterweight to Theorem 2.2, and matrix concentration (Tropp’s user-friendly tail bounds) as the probabilistic engine behind every guarantee quoted above.

Reproducibility. The test suite runs in about one second; the full experiment suite in under half a minute. Every number in Section 7 is regenerated by python examples/OX_*.py with the seeds fixed in the scripts.

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